A posteriori error estimation for SUPG–Taylor–Hood discretisations of the time-dependent Navier–Stokes equations: the AAJ27 examples in time, and an implementation of the adaptive SUPG/PSPG/grad-div method of the draft (v6)

galerkit, branch `time-navier-stokes`

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Abstract

This report continues `navier_stokes/theta_supg_th` and documents two further pieces of work on the galerkit branch `time-navier-stokes`. Part I extends Examples 5.1 and 5.2 of Afzal, Ahmed and John [2] to the time-dependent case and studies, with the SUPG–Taylor–Hood method of the previous report (no PSPG, no grad-div) and its FGJN14-type estimator (not the AAJ27 estimator), the convergence in space, the effectivity index and the contributions of the estimator. Part II is a new implementation (package `galerkit.navier_stokes.twod.adaptive_spg_th`) of the adaptive SUPG/PSPG/grad-div method of the working draft "An adaptive SUPG/PSPG/grad-div method for the time-dependent incompressible Navier–Stokes equations" (v6, November 2026) [1], for the pair P_2/P_1, together with a review of the draft, an audit of the earlier implementation, and Examples 1–3 of the draft. All rates and comparisons are *observed numerically*; nothing in this report is a proof. Claims are marked as *checked* (verified numerically or by a test), *plausible* (consistent with the data, not verified) or *open*.

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Part I
AAJ27 Examples 5.1 and 5.2 in the time-dependent case

1 Setting

The steady solutions of [2, Sect. 5] are multiplied by $g(t) = 1 + \frac{1}{2}\sin 2\pi t$, $\boldsymbol{u}(\boldsymbol{x},t) = g(t)\boldsymbol{U}(\boldsymbol{x})$, $p = g(t)P(\boldsymbol{x})$, and $\boldsymbol{f}$ is computed from the time-dependent Navier–Stokes equations: Example 5.1 is the smooth solution (35)–(36) of [2] ($\nu = 10^{-2}$ and $1/400$), Example 5.2 the solution with boundary layers at $x = 1$, $y = 1$, $\boldsymbol{u} = \operatorname{curl}\phi$, $\phi = x^2y^2(1 - e^{\lambda(x-1)})^2(1 - e^{\lambda(y-1)})^2$, $\lambda = 0.5/\sqrt{\nu}$, $p = e^{x+y} - (e-1)^2$. The coarsest mesh is that of AAJ27 (the unit square with one diagonal), levels 2–7 by red refinement (188 to 148 740 dofs), $\Delta t = 0.01$, $T = 1$, trapezoidal rule. The discretisation is the SUPG–Taylor–Hood method of the previous report (SUPG with the pressure gradient in the residual, $\delta_K = 0.25\min\{h_K/\|\boldsymbol{u}_h\|_{\infty,K}, h_K^2/\nu\}$, no PSPG, no grad-div). The estimator is the extension `supg` of FGJN14 (28) (cell residual, edge jumps, divergence term); the AAJ27 estimator (34) is *not* used, as requested.

Three error norms are reported: the AAJ27 norm $\|\cdot\|_{\text{spg,nse}}$ of [2, Sect. 5] with $\delta_K = 0.5h_K^2$, $\mu_K = 0.5$ (a yardstick: our method has no PSPG/grad-div), the SUPG norm $(\nu|e_u|_1^2 + \sum_K \delta_K\|(\boldsymbol{u}_h\cdot\nabla)e_u\|_K^2)^{1/2}$, and the *residual SUPG norm*

$$\|(e_u,e_p)\|_{\text{res}} = \Big(\nu|e_u|_1^2 + \sum_K \delta_K\|(\boldsymbol{u}_h\cdot\nabla)e_u + \nabla e_p\|_K^2\Big)^{1/2},$$

which contains the pressure where the SUPG residual, and hence the estimator, contains it.

2 Results

example	quantity	level 2	level 5	level 7	finest rate
5.1, $\nu = 10^{-2}$	res. SUPG norm	2.84	0.432	2.85e-2	1.99
	η_{supg}	18.4	2.27	0.152	1.99
	eff (res.)	6.50	5.26	5.33	
	eff (SUPG norm)	10.6	38.1	73.5	
	eff (spg, nse)	3.27	5.80	10.8	
5.1, $\nu = 1/400$	eff (res.)	6.58	5.63	5.37	
	eff (SUPG norm)	9.38	24.3	39.5	
5.2, $\nu = 10^{-2}$	res. SUPG norm	0.467	1.82e-2	1.27e-3	1.99
	eff (res.)	7.13	7.02	6.84	
	eff (SUPG norm)	16.1	21.8	36.6	

Table 1: Part I, $t = 1$, uniform meshes. The effectivity w.r.t. the residual SUPG norm varies by at most ± 0.3 between $t = 0.25, 0.5, 0.75, 1$ on every level.

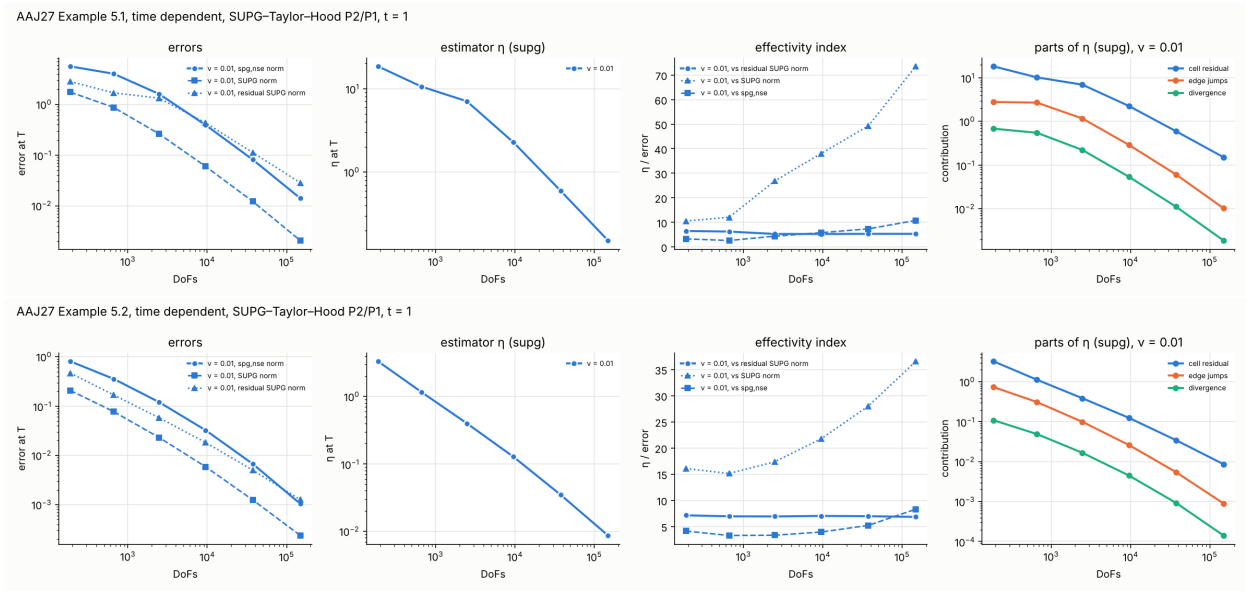


Figure 1: Part I: errors, estimator, effectivity index and contributions; Example 5.1 ($\nu = 10^{-2}$, top) and 5.2 (bottom).

Observations.

1. The effectivity of η_{supg} w.r.t. the residual SUPG norm is 5.3–7.1 on all levels, both viscosities, both examples, all output times and on the adaptive meshes (5.4–6.5) (*checked*).
2. W.r.t. the velocity-only SUPG norm it grows ($10 \to 74$): the residual contains $\nabla(p - p_h) = O(h)$ for P_1 pressure, while the SUPG norm of the velocity error is $O(h^2)$. Replacing ∇p_h by ∇p in the residual gave effectivities 6.4–9.3 (*checked*).
3. The h^1 estimator is not robust (effectivity w.r.t. the residual norm drifts from ≈ 3 to 0.7).
4. The cell residual dominates ($\geq 90\%$); edge jumps $\approx 7\%$, divergence $\approx 1\%$.
5. Adaptivity (Example 5.2, Dörfler marking with h^1 ranking and `supg` global estimate): at ≈ 9.7k dofs the energy-type errors are 1.5–1.7 times smaller than on uniform level 5, the L^2 velocity error is larger ($7.0 \cdot 10^{-4}$ vs $3.8 \cdot 10^{-4}$) (Fig. 2).

3 Example 5.2 at $\nu = 10^{-4}$ and 10^{-6}

The exact solution reaches $\max |\boldsymbol{u}| \approx 19$ ($\nu = 10^{-4}$) and ≈ 240 ($\nu = 10^{-6}$); the layer width is $1/\lambda = 0.02$ and 0.002. The degree-7 quadrature under-integrates the layers at $\nu = 10^{-6}$ (exact $|\boldsymbol{u}|_1$ by 64% on level 2 to 1.5% on level 7), so errors and estimators were evaluated with a composite rule as fine as level 8 (error $< 0.1\%$, *checked*). Newton failures were handled by halving Δt (at most twice). The runs were stopped after levels 5 ($\nu = 10^{-4}$) and 4 ($\nu = 10^{-6}$) because the results were not satisfactory:

ν	level	dofs	spg, nse	SUPG norm	η_{supg}	eff (res.)
10^{-4}	2	188	20.8	12.8	201	14.5
	5	9540	55.2	27.7	57.1	3.05
10^{-6}	2	188	29.4	20.3	24 910	1285
	4	2468	453	204	5 317	25.8

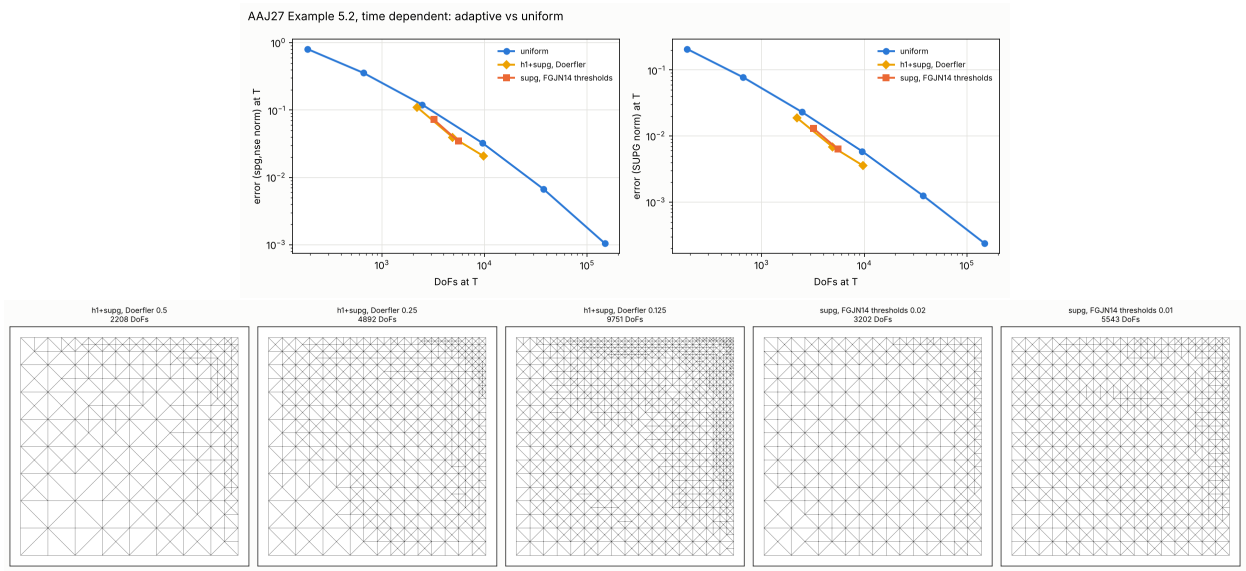


Figure 2: Part I, Example 5.2, $\nu = 10^{-2}$: adaptive vs uniform, and the adaptive meshes at $t = 1$.

All errors grow under refinement on these levels (the layers are not resolved), η decreases, and the effectivity is not robust in this pre-asymptotic regime. Plausible (not checked): the cell residual is dominated by the data $\boldsymbol{f}$ ($\approx 10^7$ near the layers at $\nu = 10^{-6}$). AAJ27 report Navier–Stokes results only for $\nu \geq 1/400$.

Part II
The adaptive SUPG/PSPG/grad-div method of the draft (v6)

4 Review of the draft

The derivations of Lemma lem:prelim and Theorem thm:upper were re-derived step by step: the Young constants ($ab \leq 2a^2 + b^2/8$, $xy \leq 10x^2 + y^2/40$), the absorption of (H0), the weights 320, 160, 640, the bound $|\Psi(\boldsymbol{e})| \leq \kappa E^2$ and the factor $(1 - 8\kappa)^{-1}$. No error was found (*checked* by hand). Points that matter for an implementation:

1. Example 2 asks for the marking of [2, Sect. 4.2] (relax until $\geq 10\%$ of the cells are marked) at every time step; this has no coarsening and no stopping rule and contradicts the threshold marking of Section 4. The threshold marking of Section 4 was used.

2. $\eta_{\mathrm{div}} = 640 h^{-2} \max_K \delta_K \|\nabla \cdot \boldsymbol{U}\|^2$ with $h = \min_K h_K$: on graded meshes the factor $h^{-2} \max \delta_K$ grows like $(h_{\max}/h_{\min})^2$ and couples all cells (observed in Sect. 9).

3. The initial data: Section 2.4 computes $(\dot{\boldsymbol{U}}^0, P^0)$ from (eq:semi) at $t = 0$. Only $\dot{\boldsymbol{U}}^0$ is used by the scheme (P^0 is only the first Newton iterate; results with $P^0 = 0$ differ by 10^{-12}, *checked*). With PSPG, $\|\nabla \cdot \boldsymbol{U}^0\| = O(h^2)$ and $\delta_K = O(h^2)$ force an $O(1)$ strong residual, so $\dot{\boldsymbol{U}}^0$ has an $O(1)$ error that does not decrease with h (*checked*: $\|\dot{\boldsymbol{U}}^0 - \partial_t \boldsymbol{u}(0)\| \approx 10$ vs $\|\partial_t \boldsymbol{u}\| = 2.35$ on levels 2–6), with consequences in Sect. 8.

4. Several statements in the draft are prophetic for the numerics: Remark rem:dtsupg(ii) (BGL07, $\Delta t \geq \alpha\delta$), Remark rem:transfer (undamped $(-1)^n$ mode after mesh changes), and the non-robustness of (H6).

5 Audit of the earlier implementation (`theta_th`)

- **Bug, fixed:** the estimator and the SUPG norm recomputed δ_K from the final velocity, while the discrete solution satisfies the auxiliary problem only with the δ_K of the step (frozen from the predictor). Now the step's δ_K is used.
- **Bug, fixed:** the indicator was evaluated with the GGS08-averaged state; it now uses the non-averaged TR solution (also in the final record when the loop is left after an averaged step).
- **Not bugs (checked):** the Galerkin pressure coupling $-(\nabla \cdot \boldsymbol{v}, p)$, $(\nabla \cdot \boldsymbol{u}, q)$, the SUPG pressure term, the component-wise loads and the SUPG mass terms agree with a direct evaluation of the forms at the quadrature points (random fields, cell-dependent parameters, mixed cell orientations) to 10^{-14} relative.

All three items are regression tests now (`tests/test_ns2d_adaptive_spg_th.py`).

6 Implementation

- **Scheme** (`scheme.py`): the forms (eq:Aspg), (eq:Lspg) with PSPG and grad-div, the TR step (eq:TR) with the discrete derivative (eq:udot) written as $R(x) = (\sigma M_t(\boldsymbol{b}) + K_t(\boldsymbol{b}))x - F(\boldsymbol{b}) - M_t(\boldsymbol{b})[H; 0]$, $\sigma = 2/\Delta t$, $H = \sigma \boldsymbol{U}^n + \dot{\boldsymbol{U}}^n$, Picard (the draft) and Newton (exact Jacobian, same fixed point), stopping on $\|R\|_2 \le 10^{-10}$ with a round-off floor $10^{-13}\|\text{rhs}\|$ (addition: the absolute test is unreachable for large right-hand sides). $\delta_K = \min\{0.5h_K^2, h_K^2/(8C_{\text{inv},K}^2\nu)\}$ (eq:deltacond), $\mu_K = 0.5$, with the inverse constant computed exactly per cell (rank-one generalised eigenproblem; $C_{\text{inv}} = \sqrt{96}$ on the AAJ27 cells with $h_K =$ diameter, so (eq:deltacond) binds at $\nu = 10^{-2}$: $\delta_K = 0.130h_K^2$). These parameters do not depend on the velocity, so "evaluating them with the AB2 predictor" is void.
- **Estimator** (`estimator.py`): η_K of (eq:etaK) with $C = 1$, $d = 2$, $\alpha = 0$, facet weights ω_F, Neumann facets, the (A1) check; optional cell-local η_{div} (not the draft, diagnostic).
- **Time control** (`timestep.py`): TR-AB2 as in Section 4 (rejection if $\|\epsilon\| > TOL_T$, factor 0.85, averaging, lower bound $\Delta t \ge \alpha \min \delta_K$, the non-averaged state for the estimator).
- **Algorithm** (`adaptive.py`): Section 4 with newest-vertex bisection, interpolation of $\boldsymbol{U}, \dot{\boldsymbol{U}}^{n+1}, \dot{\boldsymbol{U}}^n$; options (a) recompute $\dot{\boldsymbol{U}}$ from (eq:semi) after a mesh change and (b) one backward Euler step after a mesh change (Remark rem:transfer); domains: unit square, cylinder channel (polygonal cylinder, new boundary vertices projected).
- **Checks** (`checks.py`): (H0)–(H5), Θ_h, the bound (eq:upper) and the ratio of (eq:heur), with $\tilde{\boldsymbol{u}}$ from (eq:aux) on the mesh refined twice (P_2/P_1; the draft asks for a higher order pair on the same mesh) and nodal interpolation.

Verification (*checked*, 15 unit tests). Polynomial solution ($\boldsymbol{u} \in P_2$, $p \in P_1$, linear in t) reproduced to 10^{-11} including $\dot{\boldsymbol{U}}^0, P^0$; Newton Jacobian against finite differences ($< 10^{-7}$); Picard = Newton (10^{-11}); Lemma lem:exact: the TR solution satisfies (eq:disc-aux) to 10^{-11}, and the auxiliary problem solved on the same mesh returns the TR solution; the estimator vanishes for the exact discrete solution; the backward Euler step satisfies (eq:semi); forms against direct quadrature evaluation.

ν	spg error, level 2 → 6	rates	η rates	eff, levels 2–5	eff, level 6
10^{-2}	2.38, 0.769, 0.223, 5.89e-2, 2.79e-2	1.80, 1.88, 1.97, **1.09**	1.80–2.01	19.7–20.0	**10.7**
10^{-3}	4.09, 1.29, 0.415, 0.102, 2.69e-2	1.84, 1.72, 2.07, 1.95	1.81–1.98	20.3–22.7	21.2
10^{-4}	4.14, 1.31, 0.403, 0.108, 2.67e-2	1.84, 1.79, 1.95, 2.04	1.80–2.01	20.3–22.6	20.7

Table 2: Example 1, $t = 2$ (start as in the draft). The contributions are ordered $\eta_{\mathrm{res}} > \eta_\delta > \eta_{\mathrm{div}} > \eta_\mu \gg \eta_F$.

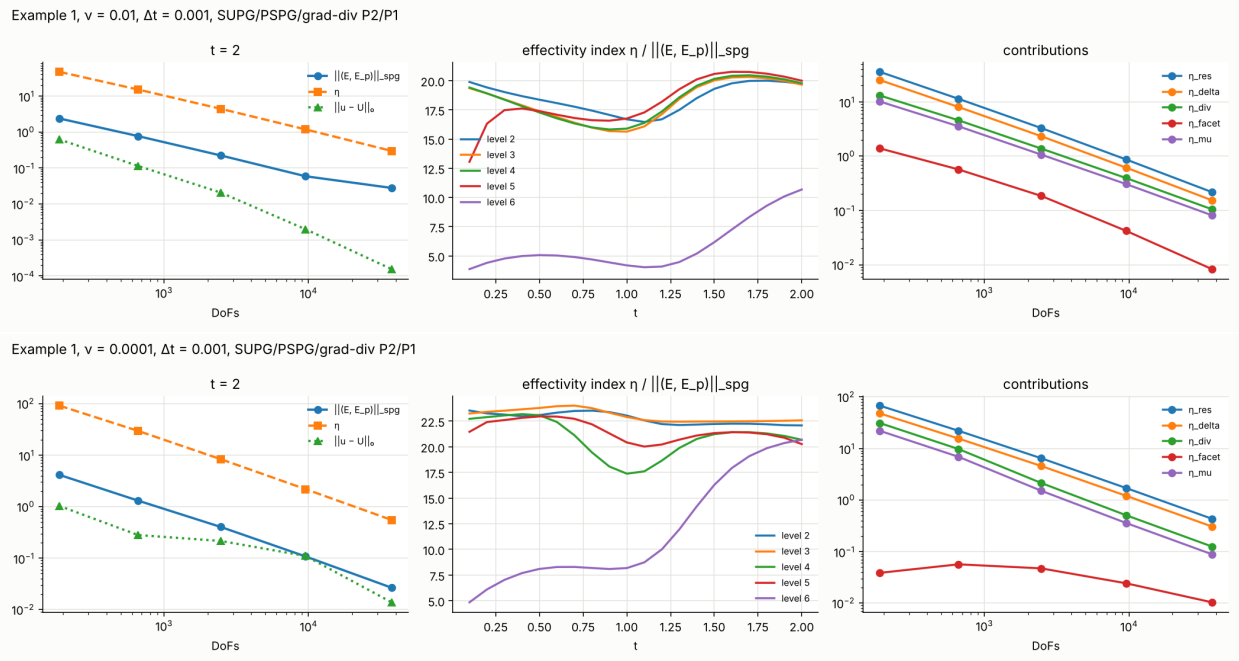


Figure 3: Example 1, $\nu = 10^{-2}$ (top) and 10^{-4} (bottom): error and estimator, effectivity versus t, contributions.

7 Example 1: smooth solution, uniform meshes

AAJ27 (35)–(36) with $(1 + \frac{1}{2}\sin \pi t)$ (velocity) and $(1 + \frac{1}{2}\cos \pi t)$ (pressure), $T = 2$, $\Delta t = 10^{-3}$, levels 2–6, start as in the draft ($\dot{\boldsymbol{U}}^0$ from (eq:semi)).

On levels 2–5 the effectivity is ≈ 20 for all ν and all t (17–24): about 2.3 times the ≈ 9 of AAJ27, consistent with the eight times larger constants of the draft ($\sqrt{8} \approx 2.8$). On level 6 it is 4–10 ($\nu = 10^{-2}$) and rises from 5 to 21 ($\nu \le 10^{-3}$) over $[0, 2]$; Sect. 8 identifies the cause.

Hypotheses. $\nu = 10^{-2}$, levels 2–5, $t = 1, 2$: (H0) $\le 5 \cdot 10^{-3}$, (H1) 0.12–0.41, (H3) ≤ 0.22, (H4) 0.2–0.75 hold; (H2) fails (2.6–6.1, required ≤ 2; only at $t = 2$ on levels 4–5 it is 1.7–1.9); (H5) holds at $t = 1$ (0.86–1.22) and fails at $t = 2$ on levels 3–5 (3.0–3.7); $\Theta_h \ll \eta$; the bound (eq:upper) holds with a large margin ($\|e\|^2/(\eta^2 + \Theta^2) \approx 0.003$); the neglected part $\|(\boldsymbol{d}, \pi)\|/\|e\| = 0.14$–$0.43$ supports (eq:heur). For $\nu \le 10^{-3}$ the steady auxiliary problem is often not solved from the TR solution (Newton and Picard fail) or a different solution is found (level 3, $\nu = 10^{-3}$: auxiliary error 9.7, larger than the full error); where solved, (H2) reaches 60 ($\nu = 10^{-4}$) and $\|(\boldsymbol{d}, \pi)\|/\|e\| = 0.6$–$0.99$, so (eq:heur) is not supported. This is consistent with the non-robustness of (H6) stated in the draft.

8 The start of the TR scheme

On level 6 the pressure error starts at 1.1–1.4 and decays slowly ($\nu = 10^{-2}$: still 0.54 at $t = 2$), and the spg error is dominated by its $\delta_K \|(\boldsymbol{U} \cdot \nabla)E + \nabla E_p\|$ part. Three starts were compared (level 6, $\nu = 10^{-2}$, $\Delta t = 10^{-3}$, $t \le 0.3$):

start	$\|p - P\|_0$, $t = 0.01/0.3$	$\|\dot{\boldsymbol{U}} - \partial_t \boldsymbol{u}\|_0$, $t = 0.3$	spg error, $t = 0.3$	effectivity
$\dot{\boldsymbol{U}}^0$ from (eq:semi) (draft)	1.45 / 1.24	7.2	5.93e-2	3.1–5.0
one backward Euler step ($\boldsymbol{U}^0$ only)	3.1e-3 / 2.6e-3	2.0e-3	1.60e-2	18.7–20.1
$\dot{\boldsymbol{U}}^0$ = interpolant of $\partial_t \boldsymbol{u}(0)$	3.1e-3 / 2.7e-3	4.0e-3	1.60e-2	18.7–20.1

The velocity error is the same in all three ($1.0 \cdot 10^{-4}$ at $t = 0.3$). The level-6 drop of the effectivity is caused entirely by $\dot{\boldsymbol{U}}^0$ from (eq:semi), whose $O(1)$ error the trapezoidal rule damps only slowly on fine meshes (*checked*). Plausible mechanism (not proved): the mode is stiff with $z \sim \Delta t/\delta_K$ (≈ 16 on level 6, 4 on level 5 at $\nu = 10^{-2}$), and $|R(z)| \to 1$. **Recommendation for the draft:** start with one backward Euler step, which needs only $\boldsymbol{u}_0$ (or take $\dot{\boldsymbol{U}}^0 = \partial_t \boldsymbol{u}_0$ if known). A rerun of Example 1 with this start was stopped after levels 2–5, whose values at $t = 2$ coincide with those of Table 2 to all printed digits.

9 Example 2: boundary layers, adaptive

AAJ27 Sect. 4.2 solution times $(1 + \frac{1}{2}\sin \pi t)$ (pressure also), $T = 1$, $TOL_T = 10^{-5}$, threshold marking ($TOL_2 = TOL_1/20$).

The algorithm as written. $\nu = 10^{-2}$ (level0 3, $TOL_1 = 0.05$) runs: ≈ 1000 cells refined at $x = 1$, $y = 1$, effectivity ≈ 22 after $t \approx 0.15$ (Fig. 4). $\nu = 10^{-3}$ blows up: the first step uses $\Delta t_0 = \min \delta_K = 0.0156$ from the inconsistent $\dot{\boldsymbol{U}}^0$; the step control then drives Δt to 10^{-7}, far below δ_K (the BGL07 regime of Remark rem:dtsupg(ii)); all cells exceed TOL_1 and the mesh doubles every step; $\dot{\boldsymbol{U}}$ grows to 10^6. (An intermediate hypothesis that η_{div} drives this runaway was refuted: the mesh refines uniformly there and both η_{div} variants coincide.)

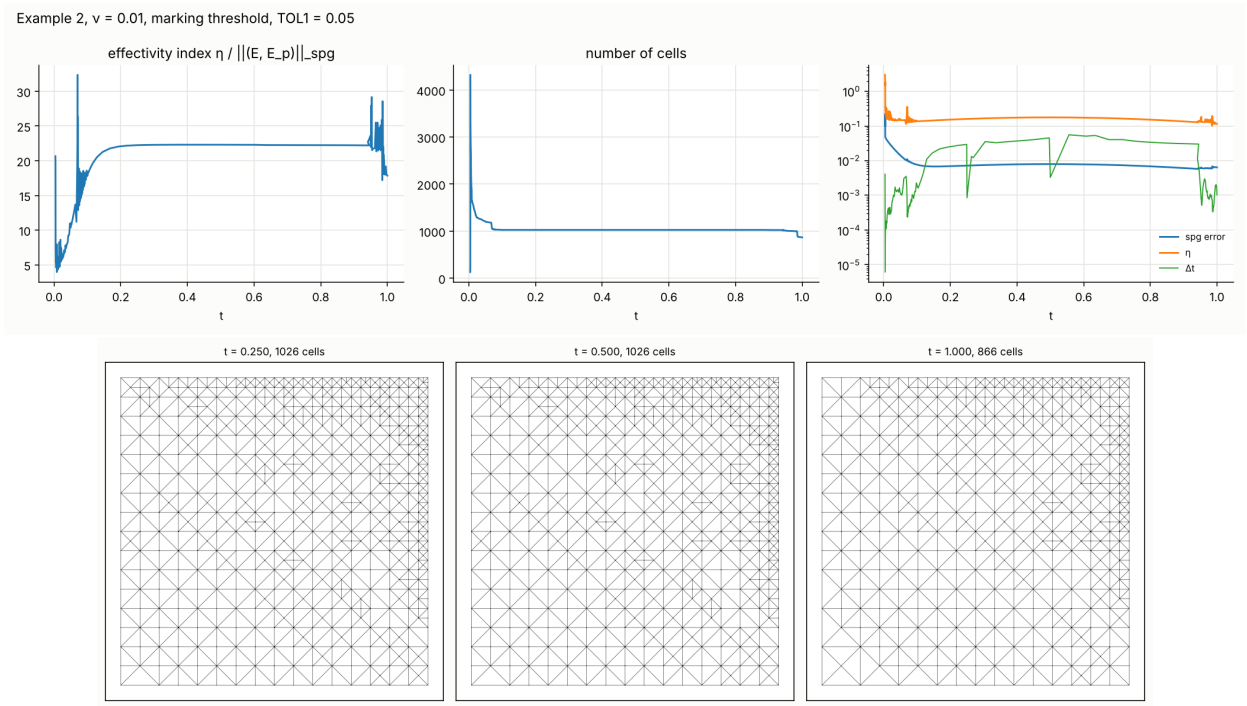


Figure 4: Example 2, $\nu = 10^{-2}$, algorithm as written: effectivity, cells, errors and step sizes; meshes.

With (a) + (b). Recompute $\dot{\boldsymbol{U}}$ from (eq:semi) after every mesh change, one backward Euler step after every mesh change, $\Delta t \geq 0.1 \min \delta_K$, initial mesh level 4 (with level 3 the first step on $h \approx 0.18$, wider than the layer 0.063, creates an error that is carried along: $\|\boldsymbol{u} - \boldsymbol{U}\|_0 \approx 0.3$–$0.45$):

run	steps (damping)	cells	eff, $t > 0.1$	spg error (T)	$\|\boldsymbol{u} - \boldsymbol{U}\|_0$ (T)
$\nu = 10^{-2}$, $TOL_1 = 0.05$	46 (2)	524	11.1–11.3	1.27e-2	1.15e-3
$\nu = 10^{-3}$, $TOL_1 = 0.3$	417 (99)	≈ 990	18.5–29.5	9.42e-2	0.145
$\nu = 10^{-3}$, cell-local η_{div} (not the draft)	315 (35)	≈ 520	7.8–12.2	0.154	0.202

Stable for both viscosities. The draft's η_{div} is $\approx 95\%$ of η on graded meshes and makes the effectivity drift with the grading; the cell-local variant gives an effectivity close to the uniform case. The L^2 velocity error grows over $[0, 1]$ at $\nu = 10^{-3}$ in both variants (*open*).

10 Example 3: flow around a cylinder

ST96 2D with inflow $1.5 \sin(\pi t/8)$, $\nu = 10^{-3}$, do-nothing outflow, $T = 8$, TR-AB2 with $TOL_T = 10^{-5}$, $\Delta t \leq 0.05$, (a)+(b), $\Delta t \geq 0.1 \min \delta_K$. The cylinder is a polygon with vertices on the circle (affine elements). Drag and lift are computed from the residual of the stabilised momentum equation tested with a function that is 1 on the cylinder.

mesh	dofs	$c_{d,\max}$ (t)	$c_{l,\max}$ (t)	$\Delta p(8)$	steps	(A1) violated
uniform level 1	5 082	2.8942 (3.918)	0.4752 (5.842)	−0.11269	914	16.3%
uniform level 2	19 668	2.9301 (3.925)	0.4145 (5.910)	−0.11352	1052	21.6%
reference [6]		2.950918 (3.93625)	0.477875 (5.69250)	−0.111616		

Drag and Δp are within 1–2%. The lift maximum is worse and later on level 2; plausible causes (not checked): Δt up to 0.03 under-resolves the shedding period ≈ 0.3, damping by grad-div ($\mu = 0.5$) and PSPG, the polygonal cylinder. (A1) is violated in 16–22% of the steps, so the upper bound of the draft does not cover them. The adaptive run (level0 1, $TOL_1 = 0.1$) failed at $t \approx 1.96$:

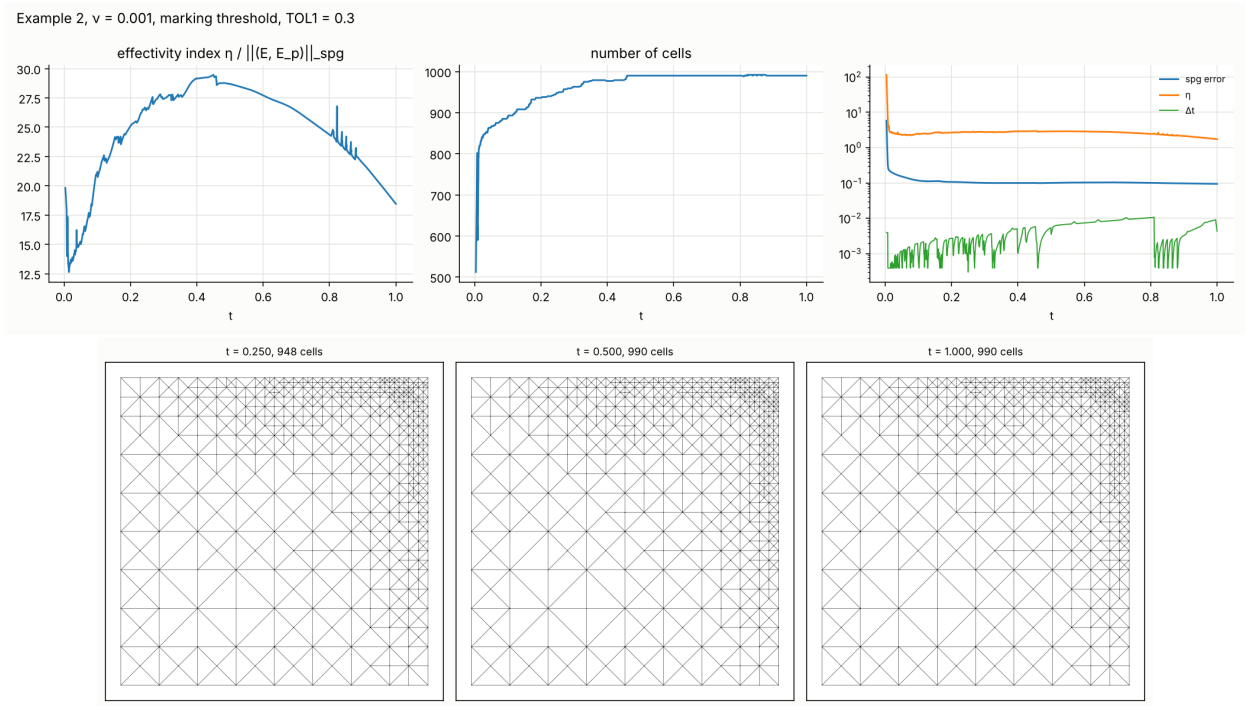


Figure 5: Example 2, $\nu = 10^{-3}$, with (a)+(b), draft η_{div}.

Δt pinned at $0.1 \min \delta_K \approx 9 \cdot 10^{-6}$ near the refined cylinder, then the mesh jumped from 3 900 to 30 000 cells and Newton diverged.

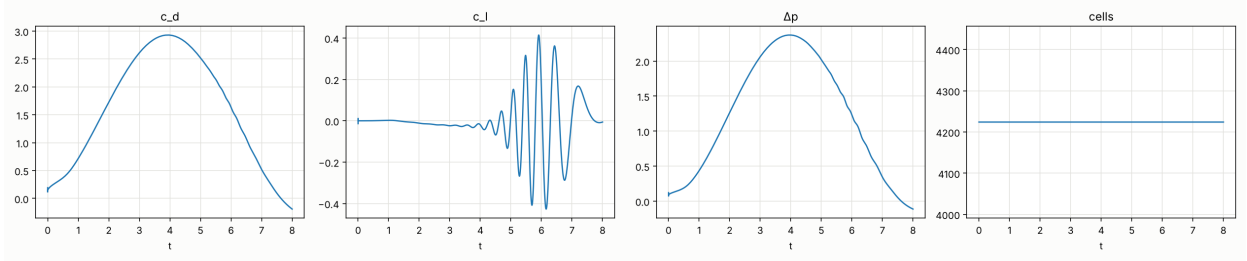


Figure 6: Example 3, uniform level 2: c_d, c_l, Δp versus t.

11 Summary

- The implementation of the draft (P2/P1) is verified (15 tests). Two bugs of the earlier implementation were fixed; the pressure coupling and the load assembly were correct.
- Example 1: the estimator is robust in h and ν on uniform meshes (effectivity ≈ 20), once the TR scheme is started with a backward Euler step instead of $\dot{\boldsymbol{U}}^0$ from (eq:semi). (H2) and, at $t = 2$, (H5) fail by moderate factors; the bound holds with a large margin. For $\nu \leq 10^{-3}$ the auxiliary steady problem is often not solvable and (eq:heur) is not supported, as anticipated by the non-robustness of (H6).
- Example 2: the algorithm as written fails at $\nu = 10^{-3}$; with (a)+(b), $\Delta t \geq 0.1 \min \delta_K$ and a sufficiently fine initial mesh it is stable. The global weight of η_{div} dominates on graded meshes.
- Example 3: uniform meshes give drag and Δp within 1–2%; the adaptive run fails.

- Suggested changes to the draft: backward Euler start; a local weight for η_{div} on graded meshes (would need a modified proof); a consistent marking for Example 2; dropping PSPG for inf-sup stable pairs (keeping SUPG and grad-div) to remove the $\Delta t \geq \alpha\delta$ conflict with the step size control.

Reproduction. galerkit branch `time-navier-stokes`:

- Part I: `examples/navier_stokes/2d/aaj27_time/aaj27_time.py`;
- Part II: `examples/navier_stokes/2d/adaptive_spg_th/example1.py`, `example2.py`, `example3.py`, with the options `--start be`, `--recompute-derivative`, `--damping`, `--dt-min-factor`, `--div-local`;
- tests: `tests/test_ns2d_adaptive_spg_th.py`.

References

[1] An adaptive SUPG/PSPG/grad-div method for the time-dependent incompressible Navier–Stokes equations, working draft, version 6, November 2026 (Sections 2–6).

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